

# EMI Shield Specification

## PRM-3A Power Regulation Module — Compartmentalized Shield Assembly

### 1. Scope

This specification defines mechanical, electrical, and environmental requirements for the electromagnetic interference (EMI) shield used in the PRM-3A Power Regulation Module Assembly. The shield provides compartmentalized isolation between the switching regulator stage and downstream analog conditioning circuitry, attenuating conducted and radiated emissions across the 30 MHz–6 GHz operating band.

This document supersedes Revision A (released 2025-11-04). Revision B updates the gasket compression specification (§4.2) and adds qualification data for the extended temperature variant (§6.3).

### 2. Mechanical Specifications

| Parameter                    | Value                   | Tolerance        | Method                      |
|------------------------------|-------------------------|------------------|-----------------------------|
| Outer dimensions (L × W × H) | 42.0 × 28.0 × 8.5 mm    | ±0.10 mm         | CMM (ASME Y14.5)            |
| Wall thickness (nominal)     | 0.40 mm                 | +0.05 / −0.02 mm | Micrometer, 12-point avg.   |
| Mass                         | 9.8 g                   | ±0.3 g           | Analytical balance          |
| Mounting flange thickness    | 0.65 mm                 | ±0.05 mm         | Micrometer                  |
| Attachment                   | 4 × M1.6 captive screws | —                | Per assembly drawing 4127-B |
| Compartment count            | 3 (RF / LDO / monitor)  | —                | Drawing 4127-B              |
| Gasket channel depth         | 0.30 mm                 | ±0.03 mm         | Profilometer                |

### 3. Electromagnetic Performance

Shielding effectiveness (SE) measured per IEEE 299-2006 in a calibrated TEM cell, with shield mounted to a representative PCB ground plane and gaskets compressed to the nominal value defined in §4.2. Measurements represent the worst-case compartment-to-compartment isolation.

| Frequency Band   | Typical SE | Minimum SE | Method                        |
|------------------|------------|------------|-------------------------------|
| 30 MHz – 100 MHz | 78 dB      | 68 dB      | IEEE 299-2006 §6.3            |
| 100 MHz – 1 GHz  | 82 dB      | 72 dB      | IEEE 299-2006 §6.4            |
| 1 GHz – 3 GHz    | 74 dB      | 62 dB      | IEEE 299-2006 §6.5            |
| 3 GHz – 6 GHz    | 65 dB      | 54 dB      | IEEE 299-2006 §6.5 (modified) |

## 4. Materials and Construction

**4.1 Shield Body.** Drawn cold-rolled steel sheet, SPCC-SD per JIS G 3141, 0.40 mm nominal thickness. Surface finish: tin plating, 3 µm minimum, per MIL-T-10727 Type I. Tin plating provides solderability for ground reflow attachment and oxidation resistance during board assembly.

**4.2 Gasket Material.** Beryllium-copper finger stock per ASTM B196, 0.10 mm thickness, electroless nickel finish. Compression range: 0.15–0.25 mm from free state. Minimum number of compression cycles before performance degradation: 10,000. Free-state height: 0.85 mm ±0.05 mm.

**4.3 Attachment Hardware.** Four (4) captive M1.6 × 4 mm steel screws with integrated lock washers, zinc-plated. Torque specification: 0.18 N·m ±0.02 N·m. Captive retention prevents screw loss during field service. Reusable up to 50 installation cycles per ISO 16047.

## 5. Environmental Requirements

| Parameter             | Specification                           | Standard                              |
|-----------------------|-----------------------------------------|---------------------------------------|
| Operating temperature | –40°C to +85°C                          | MIL-STD-810H Method 501.7 / 502.7     |
| Storage temperature   | –55°C to +125°C                         | MIL-STD-810H Method 501.7             |
| Thermal cycling       | 500 cycles (–40 to +85°C, 30 min dwell) | JESD22-A104D Condition G              |
| Humidity (operating)  | 5%–95% RH, non-condensing               | MIL-STD-810H Method 507.6             |
| Salt fog              | 96 hours, 5% NaCl @ 35°C                | ASTM B117                             |
| Vibration             | 20–2000 Hz, 14.6 Grms, 3 axes × 1 hour  | MIL-STD-810H Method 514.8 Cat. 24     |
| Mechanical shock      | 1500 G, 0.5 ms half-sine, 6 directions  | MIL-STD-810H Method 516.8             |
| Service life          | 15 years (continuous operation)         | Per MicroCo qualification plan QP-227 |

## 6. Qualification Data

**6.1 Initial Qualification.** Three (3) lots of 30 units each subjected to the full environmental test sequence. All 90 units passed all acceptance criteria. Test reports archived under QR-2473-001 through QR-2473-090.

**6.2 Production Lot Acceptance.** Each production lot (typical lot size: 500–2,000 units) is sampled per ANSI/ASQ Z1.4 Level II, AQL 0.65, with automated visual inspection on 100% of units and dimensional inspection on a sample basis. Lot acceptance criteria detailed in §7.

**6.3 Extended Temperature Variant (Rev. B addition).** A high-temperature variant (MC-SPEC-EMI-2473-B-HT) qualified for the –55°C to +125°C automotive operating range was added in Revision B. The HT variant uses an identical mechanical envelope; differences are limited to gasket material (silver-plated copper alloy substituted for the standard beryllium-copper) and plating thickness (5 µm minimum vs. the standard 3 µm). Performance data for the HT variant is documented in a separate appendix and is not included in the tables of this document.

## 7. Acceptance Criteria

Each production lot is subject to the following acceptance criteria:

**(a)** Visual: no surface defects exceeding 0.2 mm in any dimension; no plating delamination, blistering, or staining; gasket finger stock free of corrosion or damage.

**(b)** Dimensional: 100% of measured units within tolerances of §2.

**(c)** Functional: 5 units per lot subjected to abbreviated SE test (IEEE 299-2006, 100 MHz and 1 GHz only). All units must meet minimum SE values of §3.

**(d)** Documentation: each lot accompanied by Certificate of Conformance signed by MicroCo Quality Engineering, referencing this specification by document ID and revision letter.

## 8. Compliance and Standards References

| Standard      | Title                                                                                   | Application |
|---------------|-----------------------------------------------------------------------------------------|-------------|
| IEEE 299-2006 | Standard Method for Measuring the Effectiveness of Electromagnetic Shielding Enclosures | §5          |
| MIL-STD-810H  | Environmental Engineering Considerations and Laboratory Tests                           | §5          |
| MIL-T-10727   | Tin Plating, Electrodeposited or Hot-Dipped                                             | §4.1        |
| ASTM B196     | Standard Specification for Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled Bar   | §4.2        |
| ASTM B117     | Standard Practice for Operating Salt Spray (Fog) Apparatus                              | §5          |
| JESD22-A104D  | Temperature Cycling                                                                     | §5          |
| JIS G 3141    | Cold-reduced carbon steel sheet and strip                                               | §4.1        |
| ISO 16047     | Fasteners — Torque/clamp force testing                                                  | §4.3        |
| ANSI/ASQ Z1.4 | Sampling Procedures and Tables for Inspection by Attributes                             | §6.2        |
| ASME Y14.5    | Dimensioning and Tolerancing                                                            | §2          |

## 9. Revision History

| Rev. | Date       | Author      | Description                                                                             |
|------|------------|-------------|-----------------------------------------------------------------------------------------|
| A    | 2025-11-04 | J. Calderon | Initial release                                                                         |
| B    | 2026-03-15 | J. Calderon | Updated gasket compression spec (§4.2); added HT variant qualification reference (§6.3) |

| Approved By | Title                   | Signature               | Date       |
|-------------|-------------------------|-------------------------|------------|
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